
Global CO₂ Emissions: Trends, Regional Shifts, & Inequalities

1750–2024 Data Overview (Global Carbon Budget 2025, NASA)

PROJECT CODE

CLI-2026

DATE

April 2026

CONFIDENTIAL

KEY INSIGHT CO₂ concentration is accelerating rapidly, primarily driven by fossil fuels.

Latest Measurement (Feb 2026)

429 ppm

50% increase from pre-industrial baseline (~280 ppm)

Increase rate is 1.5x greater than at the end of the last ice age.

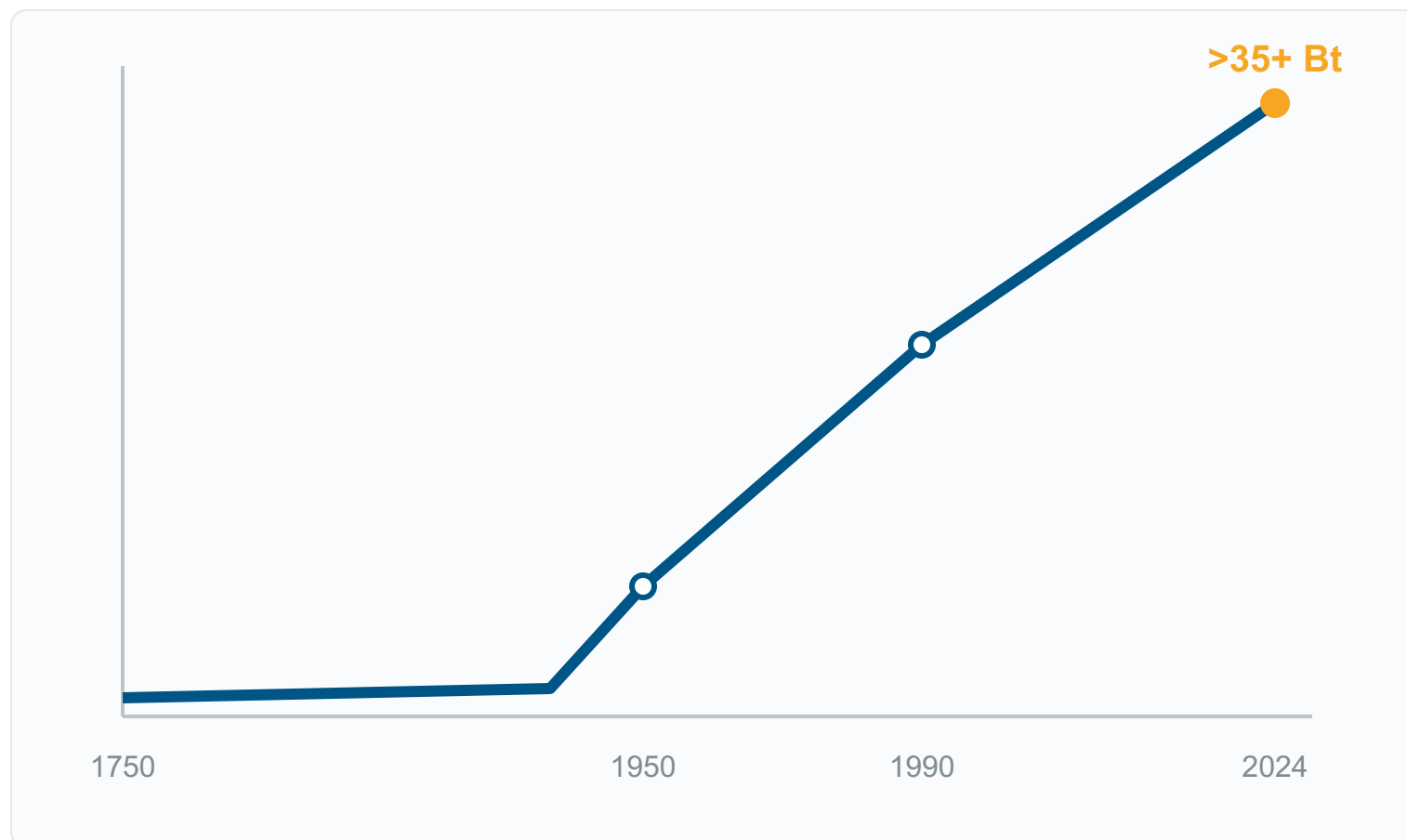


KEY INSIGHT

Emissions growth has slowed in recent years, but total volume has not yet peaked.

Historical Growth

- **1950:** ~6 Billion tonnes
- **1990:** ~20 Billion tonnes
- **Today:** >35 Billion tonnes



KEY INSIGHT

China's steep rise post-2000 contrasts with declines in the United States and Europe.

Top Emitters (Annual CO₂)

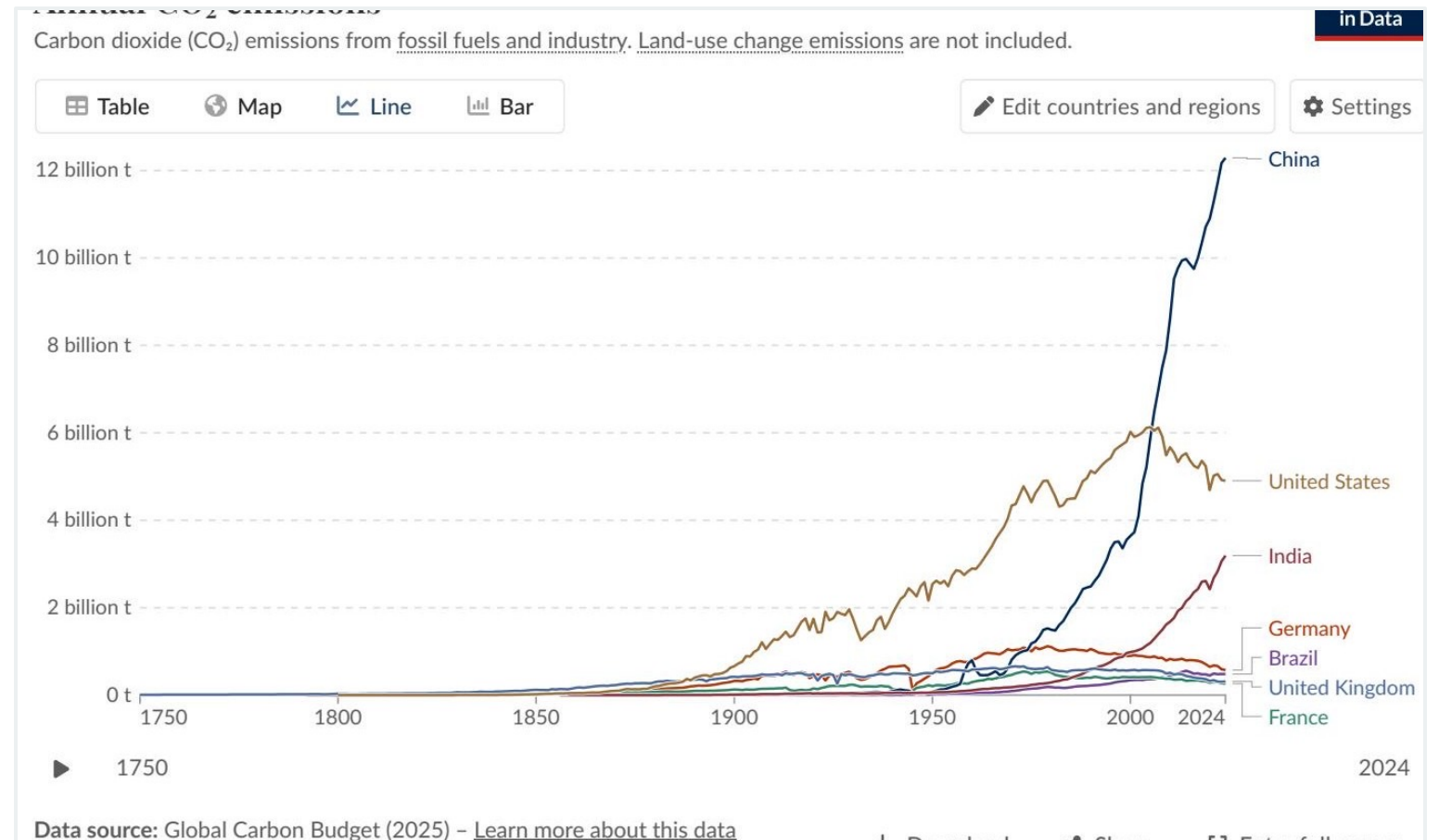
China ~12.5 Bt (27%)

United States ~5.0 Bt (14%)

India ~3.0 Bt (8%)

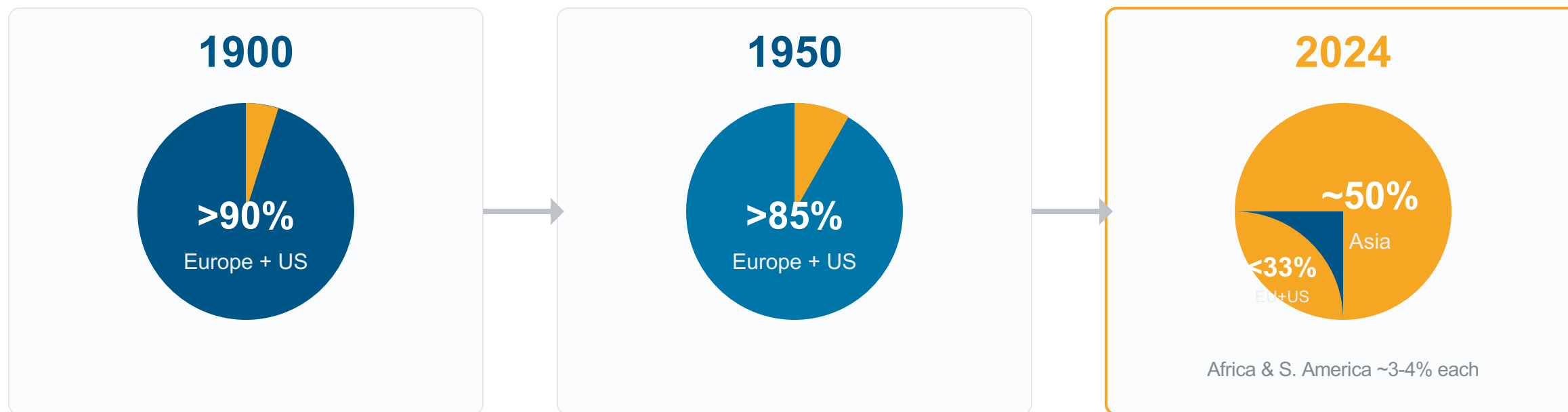
EU-27 ~2.7 Bt (7%)

China accounts for >25% of global emissions.
US emissions are declining from ~6 Bt peak.



KEY INSIGHT The center of emissions has shifted dramatically from the West to Asia over the last century.

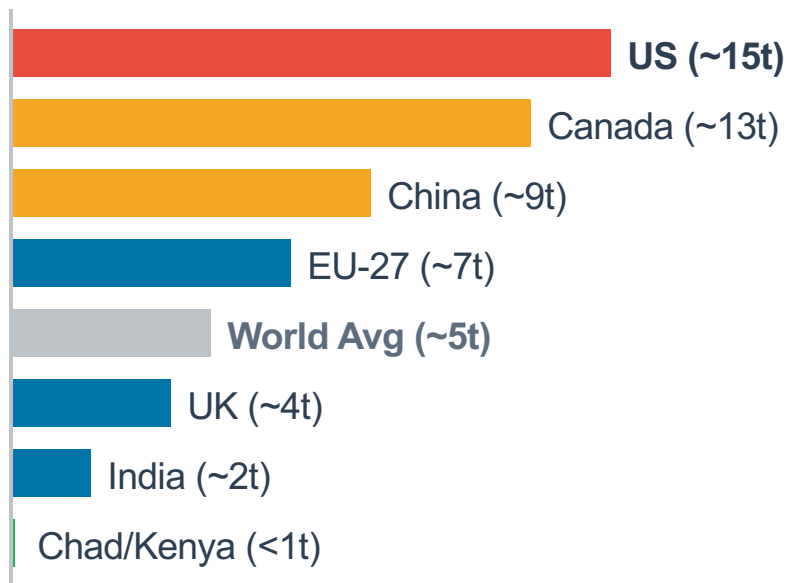
Geographic Center of Emissions Shift



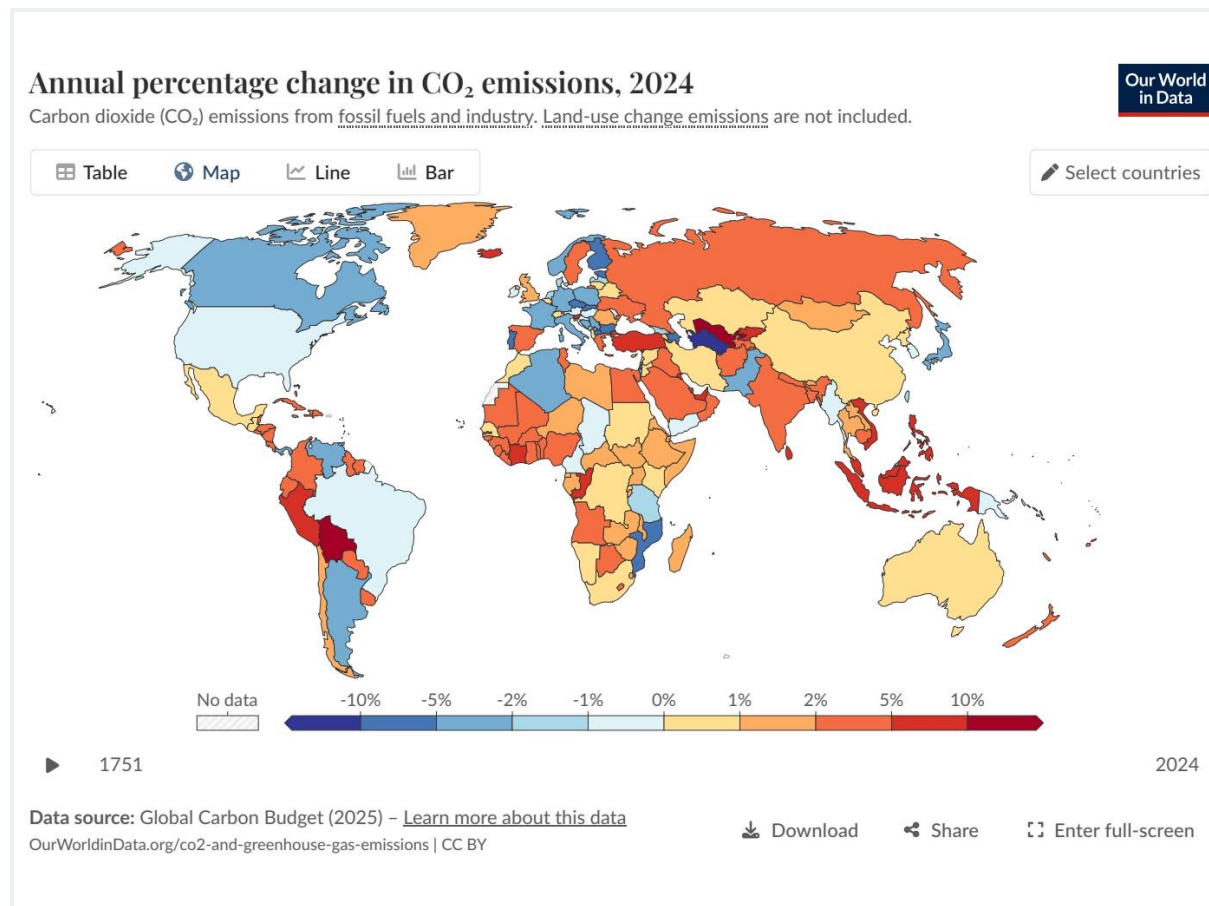
💡 KEY INSIGHT

An average American emits in two days what an average person in Chad emits in a year.

Per Capita CO₂ (tonnes/year)



150× Disparity between highest and lowest.



KEY INSIGHT The 2024 map of annual percentage change shows a deeply mixed global picture.

2024 Map Highlights



US and Europe

Recorded declines of -1% to -5%



Russia & SE Asia

Increases of +2% to +10% or more

Map reflects highly uneven annual changes globally.

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table

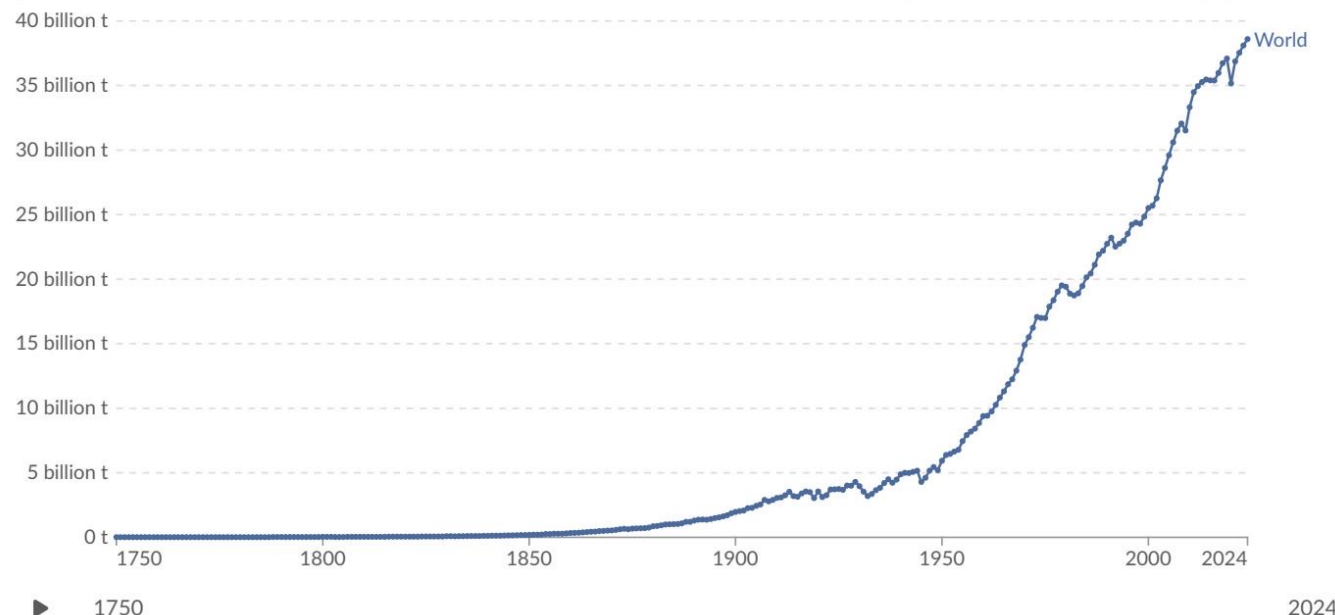
Map

Line

Bar

Edit countries and regions

Settings



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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 **KEY INSIGHT** Policy and energy mix choices determine per capita outcomes, not just prosperity.

Prosperity vs. Policy: The Energy Mix Factor

Country Group	Per Capita CO ₂	Primary Energy Mix Drivers
France & UK	~4 tonnes/yr	High nuclear (France) & renewables
Germany & Netherlands	Higher	Higher reliance on fossil fuels (e.g., Coal)

Countries with similar living standards and prosperity levels can have very different per capita emissions depending on their reliance on nuclear, renewable

 **KEY INSIGHT** Synthesizing the core findings on global CO2 emission trends.

1

Global emissions exceed 35 Bt

Growth rates have slowed, but total emissions have not yet peaked.

2

China emits >25% of global total

China's ~12.5 Bt is more than double the US; India is third and rising fast.

3

Per capita inequalities remain extreme

A 150× gap separates highest emitters (US/Aus) from the lowest (Chad/Niger).

4

US and Europe declining but historic share is large

They now account for <33% of current emissions, though historical responsibility is immense.

Policy and energy mix choices determine per capita outcomes.